



CSCE 633-700: Machine Learning

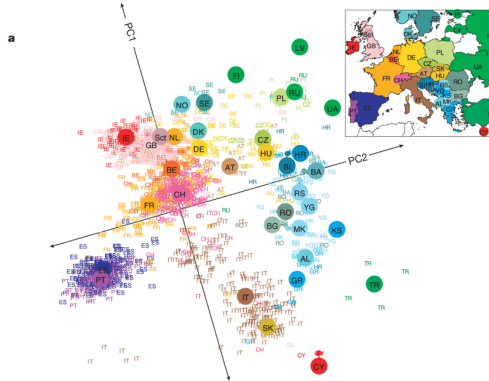
Clustering

Overview

- Clustering overview
- Partitional clustering
 - K-means clustering
 - Gaussian Mixture Models (GMM)
- Hierarchical clustering

Clustering

(1) Understanding: Finding patterns/structure/sub-populations in data



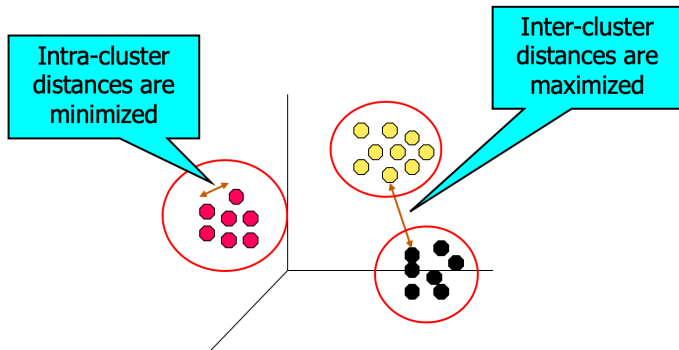
(2) Summarization: Reducing the size of large datasets

Clustering

- find patterns/structure/sub-populations in data (“knowledge discovery”)
- training data does not include desired outputs
- less well-defined problem with no obvious error metrics
- topic modeling, market segmentation, clustering of hand-written digits, news clustering (e.g. Google news)

Clustering

Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

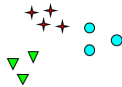


Clustering

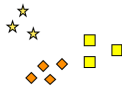
Notion of clustering can be ambiguous



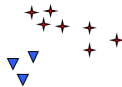
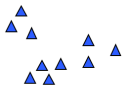
How many clusters?



Six Clusters



Two Clusters



Four Clusters

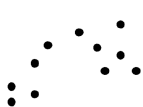


Types of clustering

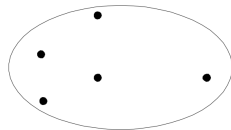
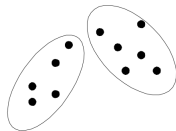
- Partitional clustering
 - non-hierarchical clusters
- Hierarchical clustering
 - a set of nested clusters organized as a hierarchical tree

Types of clustering

Partitional clustering



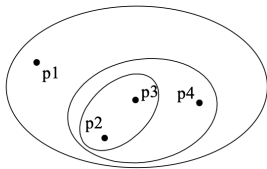
Original Points



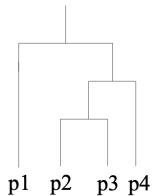
A Partitional Clustering

Types of clustering

Hierarchical clustering



Traditional Hierarchical Clustering



Traditional Dendrogram